

Task-6: Interview Questions and Answers

1. How do you group data by month and year?

You can use the `EXTRACT()` or `DATE_PART()` function in SQL to extract the year and month from a date column, then group by those values.

```
SELECT EXTRACT(YEAR FROM order_date) AS order_year, EXTRACT(MONTH FROM
order_date) AS order_month, SUM(amount) AS total_revenue FROM orders
GROUP BY order_year, order_month;
```

This groups data by both year and month.

2. What's the difference between `COUNT(*)` and `COUNT(DISTINCT col)`?

`COUNT(*)` counts all rows, including duplicates and NULLs. `COUNT(DISTINCT col)` counts only unique, non-NULL values.

Example: `COUNT(*)` counts every order; `COUNT(DISTINCT order_id)` counts each unique order only once.

3. How do you calculate monthly revenue?

By summing up the amount column grouped by month and year:

```
SELECT EXTRACT(YEAR FROM order_date) AS year, EXTRACT(MONTH FROM
order_date) AS month, SUM(amount) AS monthly_revenue FROM orders GROUP BY
year, month;
```

`SUM(amount)` gives total sales (revenue) per month.

4. What are aggregate functions in SQL?

Aggregate functions perform calculations on a set of values and return a single result.

Examples: `SUM()`, `AVG()`, `COUNT()`, `MIN()`, `MAX()`.

Used with `GROUP BY` for summaries and analytics.

5. How to handle NULLs in aggregates?

Most aggregate functions ignore NULL values automatically. You can replace them using `COALESCE()` or `IFNULL()`.

```
SELECT SUM(COALESCE(amount, 0)) FROM orders;
```

This treats NULLs as zero.

6. What's the role of `ORDER BY` and `GROUP BY`?

`GROUP BY` groups rows having the same values into summary rows (for aggregation). `ORDER BY` sorts the result set in ascending or descending order.

```
GROUP BY order_year, order_month ORDER BY order_year, order_month;
```

You group first, then sort results.

7. How do you get the top 3 months by sales?

Use ORDER BY with LIMIT (or TOP in SQL Server):

```
SELECT EXTRACT(YEAR FROM order_date) AS year, EXTRACT(MONTH FROM  
order_date) AS month, SUM(amount) AS total_revenue FROM orders GROUP BY  
year, month ORDER BY total_revenue DESC LIMIT 3;
```

This returns the top 3 months with highest revenue.